REMOTE SENSING

SCIENCE OLYMPIAD @ BERKELEY

Team Number: _____

Team Name: _____

Score: ___/78

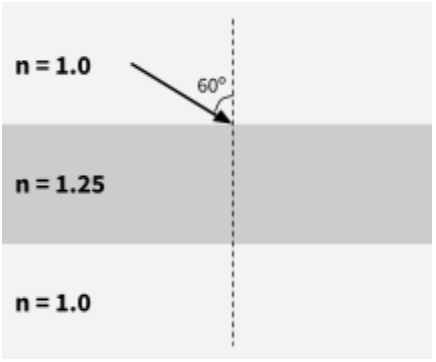
1. The blue color of the sky is primarily derived from what type of scattering? What conditions are required for this type of scattering to occur? (2)

2. Identify the most appropriate method of imaging for each of the following scenarios, choosing from: I) Active sensing II) Whisk broom III) Push broom; each term may be used multiple times or none at all. (4)

   _____ An aircraft is rapidly flying over a field and wants to image the vegetation underneath.
   _____ A satellite in geostationary orbit wants to vertically profile clouds in order to predict rainfall.
   _____ Bob wants to build his own remote sensor. He is financially limited but wants a flexible sensor that can scan various specialized geometry or terrains.
   _____ Alice also wants a remote sensor, but she wants whichever option that can provide the highest resolution.

3. What are two ways that remote sensing can be used in forecasting volcanoes? (2)

4. A beam of light enters from air (index of refraction = 1.0) to a layer of peculiar air with index of refraction of 1.25 at an angle of 60º from the normal. At what angle from the normal would the light beam emerge from the peculiar air back into air, if at all? Show your work or reasoning. (3)



Use Image 1, which depicts a region of Taiwan, for the following section.

5. What units are the margins of the map labeled with? (1)

6. What remote sensing technique is primarily used to obtain this image? Briefly explain how this technique is able to measure the elevation of the ground. (3)

7. What region of the electromagnetic spectrum does this technique primarily operate in? (1)

8. Name something else that this technique could be good for measuring. (1)

9. What is the overall trend of this region's change in elevation? (1)

a. Increase
b. Decrease
c. Constant

10. What is a likely anthropogenic cause for the change you selected in the previous question? Briefly explain the mechanism through which it occurs. (3)

11. List two natural and two anthropogenic sources of aerosols on Earth. (2)

12. Refer to Image 2, which depicts aerosol hotspots across the globe during January and July. Match the months to each image and give one reason of how you can tell. (2)

2a: _____

2b: _____

13. What instrument on Terra took these images? How does it physically measure aerosol concentration? (2)

14. Explain the Twomey effect. Be sure to explain how the addition of aerosols influences clouds and ultimately Earth's radiative balance. (3)

15. Explain how the **opposite** of what you described above might be true (e.g. if you previously concluded the Twomey effect states aerosols **cool** the Earth, now propose a process by which aerosols can **warm** the Earth, and vice versa). (2)

Images 3a, 3b and 3c display a Bay Area region in multiple filters with data from Sentinel-2. Use **Images 3a-c** to answer the following section (please hold off on referring to **3d and 3e** until further instructed).

16. How is false color implemented in the context of remote sensing (as opposed to true color)? Which image(s) between 3a-c display(s) false coloration, if any? (3)

17. What orbit is Sentinel-2 in and why might that be helpful for its task? (2)

18. Sentinel-2 has a spatial resolution of 10 meters in Image 3a. A square of what area is covered by each pixel? Assuming the images provided are 32.5 km by 53 km, what is the overall resolution of the Image 3a? Show your work. (4)

19. Match the following descriptions of Sentinel-2 to the type of resolution being described, choosing from: Temporal, Radiometric, Spectral. (3)

_____ Sentinel-2's 12-bit resolution allows it to distinguish between 4,096 levels of brightness

_____ Sentinel-2 has 13 bands, covering the electromagnetic spectrum from 442 nm to 2202.4 nm

_____ Sentinel-2 revisits the same location every 5 days

20. What general region of the electromagnetic spectrum is used to produce Image 3a? (1)

21. Are the clouds displayed near the top right of Image 3a high or low clouds? Do you expect them to contribute to a net warming or cooling of the surface? Explain. (3)

22. In Image 3a, what color generally corresponds to the densest vegetation? (1)
    a. Brown
    b. Yellow
    c. White
    d. Blue
    e. Green

23. In Image 3b, what color generally corresponds to the densest vegetation? (1)
    a. Red
    b. Yellow
    c. Orange
    d. Light green
    e. Dark green

24. In Image 3c, what generally color corresponds to the densest vegetation? (1)
    a. White
    b. Blue
    c. Light green
    d. Dark green

25. Which of the labeled points on the surface has the highest albedo? (1)
    a. A
    b. B
    c. C
    d. D

26. Place the following electromagnetic regions on the spectrum provided: near infrared (NIR), far infrared (FIR), short-wave infrared (SWIR), mid-wave infrared (MWIR), long-wave infrared (LWIR). (2)

Visible _____ _____ _____ _____ _____ Microwave

27. Image 3b utilizes the NDVI to display vegetation health. What does NDVI stand for? (1)

28. The NDVI calculation requires the visible red and what region of the infrared spectrum as listed in the previous question? What bands do they correspond to, respectively? (2)

29. The data utilized in NDVI primarily relies on which of the following interactions of light with vegetation? (1)

a. Reflection
b. Refraction
c. Diffraction
d. Scattering

30. Image 3c utilizes the NDWI, a similar index for water instead of vegetation. It can be calculated using the following formula for Sentinel-2: NDWI = (Band 3 - Band 8) / (Band 3 + Band 8). Circle the words that best complete the following sentence: (2)

If Band 3 receives a much stronger signal than Band 8, the NDWI will be **(low/high)**, indicating **(water/dry land)**. Band 8 is **(longer/shorter)** in wavelength than Band 3, as water produces stronger signal in **(longer/shorter)** wavelengths compared to dry land.

31. What is the range of all possible values for the NDWI? In other words, what are the lowest and highest possible NDWI values? (1)

32. What color in image 3c has the lowest NDWI value? (1)

33. Which of the following best identifies the features circled in E on the maps? (1)
    a. Greenbelts
    b. Buildings
    c. Crops field
    d. Artificial ponds

34. Use evidence from at least 2 images between 3a-c to support your choice above. (2)

35. Other than point B, which labeled point/region would you expect to have the highest risk of flooding (choose from A, C, D, E)? Briefly explain your answer. (2)

36. At which labeled point would you expect to have the lowest variation in daily temperature? Why? (2)

37. Notice the main river on the left. Which general direction is it flowing in? (1)
    a. Northeast
    b. Northwest
    c. Southeast
    d. Southwest

38. Notice that the majority of the branches of the river are to the west side of the river. Why might that be the case? (2)

39. Refer to image Image 4d, which shows the annual NLCD land classification by the USGS in 2024. Assign a Land Cover Class, as below, to each labeled point in Image 4d. (4)

a. Developed
b. Barren
c. Herbaceous
d. Shrubland
e. Cultivated
f. Wetland

____ Point W
____ Point X
____ Point Y
____ Point Z

40. Compare Image 4d with 4e, which is the NLCD classification for 2004. What are two differences that you notice? (2)

Tiebreaker: What is the CO2 level today (as seen on co2.earth)? Provide your answer in ppm to at least two decimal places. (0)